

Course 6044C

Semester VI

Theory

4 Credits

Drone Technology

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes

Electronics Engineering

Contact hours

60

Teaching scheme

3:1:0:0

Credits

4

CIE / ESE

Theory CIE 40 / Theory ESE 60

Self-learning

5.5 h/week

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6044C as Drone Technology in Semester VI for Electronics Engineering. The official SITTTTR detailed course page was checked at the indexed link and returned "Requested file not found." With the user's approved public-source approach, this handbook is transparently based on NPTEL IISc Bangalore's Drone Systems and Control course and polytechnic drone technology curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Drone designs, flight plans, pilot operations and maintenance procedures must be based on approved engineering plans, certified aviation data, current DGCA regulations and competent professional review.

Verified source note: Verified sources support UAV classifications, aerodynamics, drone components, and DGCA rules. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60

Contact hours

4

Credits

Theory CIE 40

CIE

Theory ESE 60

ESE

Course introduction

Drone Technology (Unmanned Aerial Vehicles - UAV) studies the design, operation and applications of pilotless aircraft systems. It develops the ability to analyze flight aerodynamics

and dynamics, evaluate drone components like flight controllers and sensors, understand DGCA regulations and flight planning, and coordinate drone-based solutions while respecting the technical, legal and safety boundaries of electronic and aerospace engineering.

Malayalam support: ു handbook ു syllabus topic ു; ു principle, diagram/procedure, application, common mistake ു ു.

Prerequisite foundation

Basic electronics, microcontrollers, sensors, wireless communication and basic physics (mechanics).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 • Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 • Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 • Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 • Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain the fundamentals of UAVs, drone classifications and basic aerodynamics.
2. Analyze the hardware components of a drone including flight controllers, motors and sensors.
3. Evaluate the flight dynamics, control systems and wireless communication for drones.
4. Explain the systematic DGCA regulations, flight safety and maintenance of drone systems.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

Malayalam support: CO ഉള്ളത് exam-ൽ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ ഉപയോഗിക്കാൻ. Define, explain, apply ഉപയോഗിക്കാൻ action words ഉപയോഗിക്കാൻ.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain the fundamentals of UAVs, aerodynamics and drone classifications.	Understand
CO2	Analyze the selection and operation of drone hardware components.	Apply
CO3	Analyze flight dynamics, control algorithms and communication links.	Apply
CO4	Explain DGCA regulations, flight safety protocols and drone maintenance.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7-PO11
CO1	3	2	2	2	2	-	-
CO2	3	2	2	2	2	-	-
CO3	3	2	2	2	2	-	-
CO4	3	2	2	2	2	-	-

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters

Module	Title	Official major topics	Hours	CO	Handbook section
1	Introduction to UAVs and Aerodynamics	Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.	Approx. 14 hours	CO1	Module 1
2	Drone Hardware and Components	Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.	Approx. 16 hours	CO2	Module 2
3	Flight Dynamics and Control	6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.	Approx. 16 hours	CO3	Module 3
4	Regulations, Safety and Maintenance	DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Introduction to UAVs and Aerodynamics

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

CO1 · Approx. 14 hours

Drone Hardware and Components

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

CO2 · Approx. 16 hours

Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

CO3 · Approx. 16 hours

Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

CO4 · Approx. 14 hours

MODULE 1

Introduction to UAVs and Aerodynamics

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

Approx. 14 hours

CO1

Understand · Apply

Learning goals

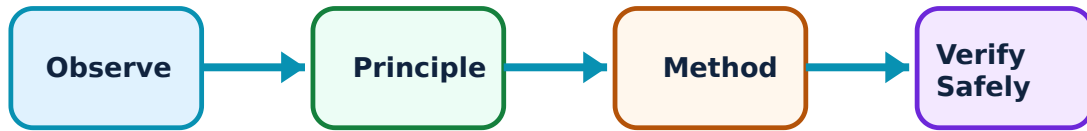
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Introduction to UAVs and Aerodynamics: identify the system, state its principle, follow the correct method and verify the result safely.

1. Drone Fundamentals–

A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board. Drones are classified by weight (e.g., Nano < 250g) and configuration. Multi-rotors (Quadcopters) are popular for their VTOL (Vertical Take-Off and Landing) capability, while Fixed-wing drones offer better efficiency for long-range tasks.

Study and application method

List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കണ്ടെത്തുക concept പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക. പട്ടിക diagram / circuit / formula / procedure പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക. പട്ടികപ്പെടുത്തുക result പട്ടികപ്പെടുത്തുക application പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക. Technical terms English-ൽ പട്ടികപ്പെടുത്തുക പട്ടികപ്പെടുത്തുക.

Common mistake / precaution: Improper classification can lead to legal violations. Do not operate drones without authorised understanding of weight limits and corresponding regulatory requirements.

2. Aerodynamics and Flight Basics–

Lift is generated by pressure differences (Bernoulli's principle). Weight is the force of gravity. Thrust is provided by motors/propellers. Drag is air resistance. Stable flight requires a balance of these forces. Control is achieved by varying motor speeds in multi-rotors.

Study and application method

Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure ഉപയോഗിക്കണം. ഏതു result ലഭിക്കും application എന്തായിരിക്കും. Technical terms English-ൽ എങ്ങനെ എഴുതണം.

Common mistake / precaution: Aerodynamic instability can cause crashes. Do not fly in high-wind conditions without authorised analysis of the drone's wind-resistance rating.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Drone Hardware and Components

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

Approx. 16 hours

CO2

Understand · Apply

Learning goals

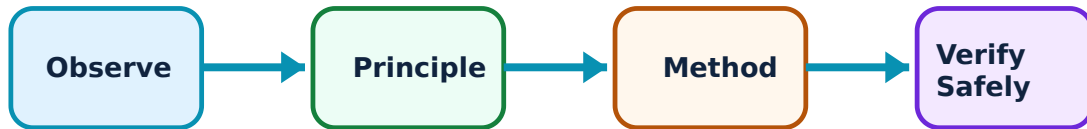
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Drone Hardware and Components: identify the system, state its principle, follow the correct method and verify the result safely.

1. Flight Controller and Sensors–

The Flight Controller (FC) is the 'brain' of the drone. It uses an IMU (Inertial Measurement Unit) containing accelerometers and gyroscopes to maintain stability. GPS provides position data, and a Barometer measures altitude. The FC processes sensor data to send commands to the ESCs.

Study and application method

Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. diagram / circuit / formula / procedure വ്യക്തമായി വിശദീകരിക്കുക. വോൾട്ട്, കറന്റ്, ഫ്രീക്വൻസി, ടൈം എന്നിവയുടെ യൂണിറ്റ് ഉൾപ്പെടെയുള്ളവ. result വ്യക്തമായി വിശദീകരിക്കുക. application വ്യക്തമായി വിശദീകരിക്കുക. Technical terms English-ൽ വ്യക്തമായി വിശദീകരിക്കുക.

Common mistake / precaution: Sensor calibration is critical. Do not attempt flight without authorised 'Level' and 'Compass' calibration to prevent erratic behavior or 'flyaways'.

2. Propulsion and Power–

BLDC (Brushless DC) motors provide high power-to-weight ratios. ESCs control motor speed. LiPo (Lithium Polymer) batteries are preferred for their high energy density. Propeller selection (diameter and pitch) must match the motor and payload requirements.

Study and application method

Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: LiPo batteries are a fire hazard if damaged or overcharged. All battery handling must follow authorised charging, storage, and disposal protocols.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

Approx. 16 hours

CO3

Understand · Apply

Learning goals

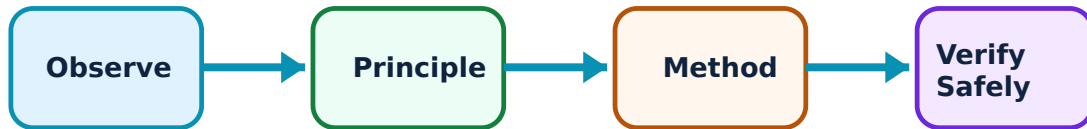
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Flight Dynamics and Control: identify the system, state its principle, follow the correct method and verify the result safely.

1. Flight Dynamics and PID Control–

Drones operate in 6 Degrees of Freedom (6-DOF). PID (Proportional-Integral-Derivative) controllers are used to maintain the desired attitude by adjusting motor speeds based on error signals. Tuning PID values is essential for smooth and stable flight.

Study and application method

Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept നൽകിയിരിക്കുന്നു. ഏതു diagram / circuit / formula / procedure നൽകിയിരിക്കുന്നു. ഏതു result നൽകിയിരിക്കുന്നു application നൽകിയിരിക്കുന്നു. Technical terms English-ൽ എഴുതിക്കുക.

Common mistake / precaution: Incorrect PID values can cause oscillations or crashes. Do not perform high-speed maneuvers without authorised verification of control stability.

2. Communication and FPV–

Radio control (RC) links allow the pilot to command the drone. Telemetry provides real-time data (battery, altitude) back to the Ground Control Station (GCS). FPV systems use a camera and video transmitter (VTX) to provide a pilot's-eye view.

Study and application method

Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result ന്നുള്ള application ന്നുള്ള Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Signal loss can lead to loss of control. All flights must follow authorised 'Fail-safe' configurations (e.g., Return-to-Home) in case of RC signal loss.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

Approx. 14 hours

CO4

Understand · Apply

Learning goals

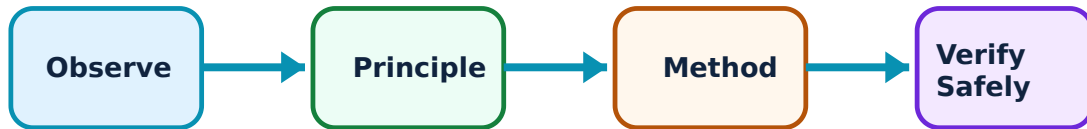
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Regulations, Safety and Maintenance: identify the system, state its principle, follow the correct method and verify the result safely.

1. DGCA Regulations and Licensing–

In India, the DGCA regulates drone operations via the 'Digital Sky' platform. Drones (except Nano) require a Unique Identification Number (UIN). Pilots must have a Remote Pilot License for commercial operations. Operations are restricted in 'Yellow' and 'Red' zones.

Study and application method

Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: കോൺസെപ്റ്റ് വിശദീകരിക്കുക. ഫോർമുല / സർക്യൂട്ട് / പ്രോസീഡർ വിശദീകരിക്കുക. ഫലം പ്രയോജനപ്പെടുത്തുക. Technical terms English-ൽ ഉപയോഗിക്കുക.

Common mistake / precaution: Flying in 'No-Fly Zones' (Red zones) is a serious offense. Always check the authorised airspace map on Digital Sky before every flight.

2. Safety and Maintenance–

Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS). Regular maintenance involves inspecting motors, cleaning sensors, and checking for frame cracks. Troubleshooting involves analyzing flight logs.

Study and application method

Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: ഏതു concept ന്നുള്ള diagram / circuit / formula / procedure ന്നുള്ള result application ന്നുള്ള technical terms English-ൽ എഴുതുക. Technical terms English-ൽ എഴുതുക.

Common mistake / precaution: Mechanical failure in flight can cause injury or property damage. Do not skip pre-flight inspections; ensure all maintenance follows authorised manufacturer guidelines.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Introduction to UAVs and Aerodynamics

Use the concept flow to revise observation, principle, method and verification.

Module 2: Drone Hardware and Components

Use the concept flow to revise observation, principle, method and verification.

Module 3: Flight Dynamics and Control

Use the concept flow to revise observation, principle, method and verification.

Module 4: Regulations, Safety and Maintenance

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare Multi-rotor and Fixed-wing drones in a conceptual table showing three differences.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch the 'Motor Layout' and rotation directions for a standard X-configuration Quadcopter.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw the 'Airspace Zone' diagram showing Green, Yellow, and Red zones conceptually.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the 'Flight Time' conceptually for a given battery capacity and average current draw.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify a drone application in your environment (e.g., photography, delivery, agriculture) and note its benefits.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Introduction to UAVs and Aerodynamics

Explain Drone Fundamentals and state one correct method, application or precaution.

A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board. Drones are classified by weight (e.g., Nano < 250g) and configuration. Multi-rotors (Quadcopters) are popular for their VTOL (Vertical Take-Off and Landing) capability, while Fixed-wing drones offer better efficiency for long-range tasks.

Answer framework: List the 'DGCA Weight Classifications' for drones; explain the four 'Aerodynamic Forces' acting on a drone in flight.

Important point: Improper classification can lead to legal violations. Do not operate drones without authorised understanding of weight limits and corresponding regulatory requirements.

Explain Aerodynamics and Flight Basics and state one correct method, application or precaution.

Lift is generated by pressure differences (Bernoulli's principle). Weight is the force of gravity. Thrust is provided by motors/propellers. Drag is air resistance. Stable flight requires a balance of these forces. Control is achieved by varying motor speeds in multi-rotors.

Answer framework: Describe how a 'Quadcopter' achieves 'Yaw' and 'Pitch' movements conceptually; compare 'Multi-rotor' and 'Fixed-wing' drones.

Important point: Aerodynamic instability can cause crashes. Do not fly in high-wind conditions without authorised analysis of the drone's wind-resistance rating.

Module 2 — Drone Hardware and Components

Explain Flight Controller and Sensors and state one correct method, application or precaution.

The Flight Controller (FC) is the 'brain' of the drone. It uses an IMU (Inertial Measurement Unit) containing accelerometers and gyroscopes to maintain stability. GPS provides position data, and a Barometer measures altitude. The FC processes sensor data to send commands to the ESCs.

Answer framework: Sketch the 'Block Diagram' of a drone's electronic system; explain the role of the 'IMU' in flight stabilization.

Important point: Sensor calibration is critical. Do not attempt flight without authorised 'Level' and 'Compass' calibration to prevent erratic behavior or 'flyaways'.

Explain Propulsion and Power and state one correct method, application or precaution. –

BLDC (Brushless DC) motors provide high power-to-weight ratios. ESCs control motor speed. LiPo (Lithium Polymer) batteries are preferred for their high energy density. Propeller selection (diameter and pitch) must match the motor and payload requirements.

Answer framework: Explain the function of an 'Electronic Speed Controller' (ESC); list three safety precautions for handling 'LiPo Batteries'.

Important point: LiPo batteries are a fire hazard if damaged or overcharged. All battery handling must follow authorised charging, storage, and disposal protocols.

Module 3 — Flight Dynamics and Control

Explain Flight Dynamics and PID Control and state one correct method, application or precaution. –

Drones operate in 6 Degrees of Freedom (6-DOF). PID (Proportional-Integral-Derivative) controllers are used to maintain the desired attitude by adjusting motor speeds based on error signals. Tuning PID values is essential for smooth and stable flight.

Answer framework: Define 'Roll', 'Pitch', and 'Yaw' in drone dynamics; explain the purpose of 'PID Tuning' conceptually.

Important point: Incorrect PID values can cause oscillations or crashes. Do not perform high-speed maneuvers without authorised verification of control stability.

Explain Communication and FPV and state one correct method, application or precaution. –

Radio control (RC) links allow the pilot to command the drone. Telemetry provides real-time data (battery, altitude) back to the Ground Control Station (GCS). FPV systems use a camera and video transmitter (VTX) to provide a pilot's-eye view.

Answer framework: Describe the role of 'Telemetry' in drone operations; explain the concept of 'First Person View' (FPV).

Important point: Signal loss can lead to loss of control. All flights must follow authorised 'Fail-safe' configurations (e.g., Return-to-Home) in case of RC signal loss.

Module 4 — Regulations, Safety and Maintenance

Explain DGCA Regulations and Licensing and state one correct method, application or precaution. –

In India, the DGCA regulates drone operations via the 'Digital Sky' platform. Drones (except Nano) require a Unique Identification Number (UIN). Pilots must have a Remote Pilot License for commercial operations. Operations are restricted in 'Yellow' and 'Red' zones.

Answer framework: Explain the 'Digital Sky' platform conceptually; list the requirements for obtaining a 'Remote Pilot License'.

Important point: Flying in 'No-Fly Zones' (Red zones) is a serious offense. Always check the authorised airspace map on Digital Sky before every flight.

Explain Safety and Maintenance and state one correct method, application or precaution. –

Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS). Regular maintenance involves inspecting motors, cleaning sensors, and checking for frame cracks. Troubleshooting involves analyzing flight logs.

Answer framework: Develop a 'Pre-flight Checklist' conceptually; list three common drone maintenance tasks.

Important point: Mechanical failure in flight can cause injury or property damage. Do not skip pre-flight inspections; ensure all maintenance follows authorised manufacturer guidelines.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Introduction to UAVs and Aerodynamics.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Drone Hardware and Components.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Flight Dynamics and Control.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Regulations, Safety and Maintenance.

8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing..
2. Develop a complete answer or practical plan for: Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials..
3. Develop a complete answer or practical plan for: 6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems..
4. Develop a complete answer or practical plan for: DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting..

LAST-MILE REVISION

Quick Revision

Module 1: Introduction to UAVs and Aerodynamics

Definition of UAV/Drone; History and Evolution; Classifications: Nano, Micro, Small, Medium, Large; Aerodynamics: Lift, Weight, Thrust, Drag; Multi-rotor vs Fixed-wing.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Drone Hardware and Components

Flight Controller (FC): Architecture and Sensors (IMU, Barometer, GPS); Electronic Speed Controllers (ESC); BLDC Motors and Propellers; Battery technology (LiPo); Frame materials.

- Match each topic to CO2.

- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Flight Dynamics and Control

6-DOF Dynamics: Roll, Pitch, Yaw; Control algorithms: PID control basics; Wireless communication: Radio control (RC), Telemetry links; FPV (First Person View) systems.

- Match each topic to C03.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 4: Regulations, Safety and Maintenance

DGCA Regulations: Digital Sky platform, UIN, Remote Pilot License; Airspace zones: Green, Yellow, Red; Pre-flight and Post-flight checklists; Maintenance and Troubleshooting.

- Match each topic to C04.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus

Term / topic	One-line meaning
Drone Fundamentals	A Drone or Unmanned Aerial Vehicle (UAV) is an aircraft without a human pilot on board.
Aerodynamics and Flight Basics	Lift is generated by pressure differences (Bernoulli's principle).
Flight Controller and Sensors	The Flight Controller (FC) is the 'brain' of the drone.
Propulsion and Power	BLDC (Brushless DC) motors provide high power-to-weight ratios.
Flight Dynamics and PID Control	Drones operate in 6 Degrees of Freedom (6-DOF).
Communication and FPV	Radio control (RC) links allow the pilot to command the drone.
DGCA Regulations and Licensing	In India, the DGCA regulates drone operations via the 'Digital Sky' platform.
Safety and Maintenance	Safety protocols include pre-flight checks (battery, propellers, GPS lock) and maintaining 'Visual Line of Sight' (VLOS).

LEARNING RESOURCES

References

Recommended drone technology references

1. Terry Kilby and Belinda Kilby, Getting Started with Drones.
2. Baichtal John, Building Your Own Drones: A Beginner's Guide to Drones, UAVs, and ROVs.
3. K. P. Valavanis and G. J. Vachtsevanos, Handbook of Unmanned Aerial Vehicles.
4. DGCA India, Drone Rules 2021 and Amendments.

Approved public drone technology sources

- <https://nptel.ac.in/courses/101108661>
- https://onlinecourses.nptel.ac.in/noc26_ae27/preview

These sources provide the verified study basis while official Course 6044C detail is unavailable; they do not replace official aviation regulations, authorised pilot training, certified equipment specifications or authorised strategic aerial plans.